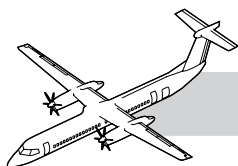


CHAPTER 54

NACELLE

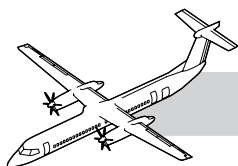
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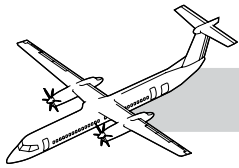
CHAPTER 54 NACELLE



54-00-00 INTRODUCTION

Three main areas of structure comprise the nacelle:

- Forward
- Center
- Aft.



GENERAL

Refer to Figure 54-1. Forward Nacelle Structure.

The forward nacelle structure is located between nacelle stations -15.95 and +133.020 and includes the following systems/components:

- Engine mount system
- Frames
- Fairings
- Engine access doors
- Lower cowl.

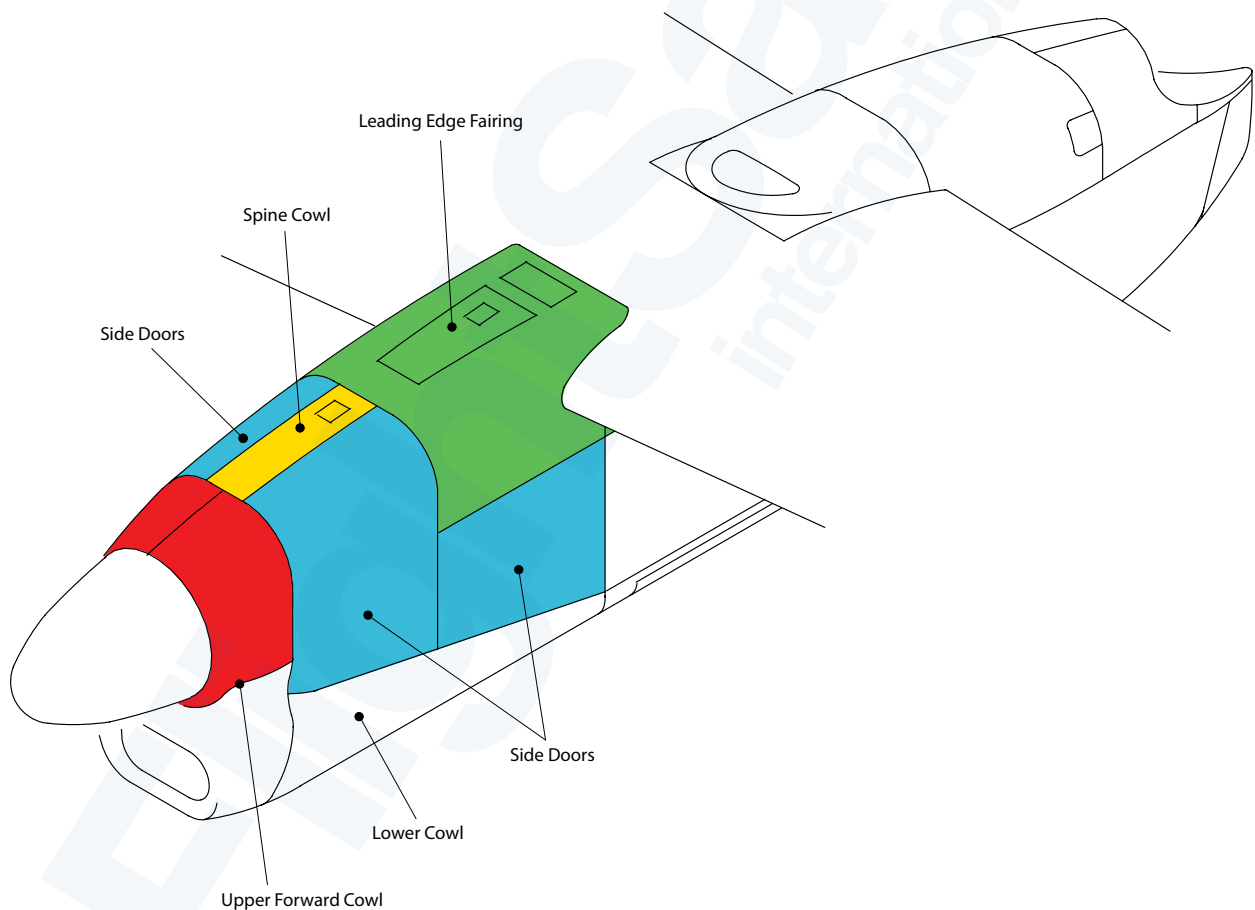
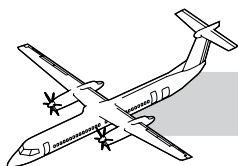
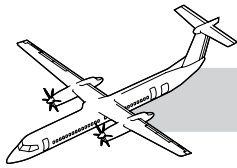


Figure 54-1. Forward Nacelle Structure



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SYSTEM DESCRIPTION

Refer to Figure 54-2. Forward Nacelle Structure, Frame and Struts.

The engine mount system struts are stainless steel tubes with stainless steel end fittings swaged in place. The struts join the forward-frame to the mid-frame, and the mid-frame to the firewall bulkhead.

There are two frames in the forward nacelle. The forward frame is a machined 6AL-4V titanium component. Its primary purpose is to attach the forward engine isolators and to distribute the engine loading to the forward struts. Titanium 6AL-4V has advantages in strength, weight and resistance to temperature, however, it is notch sensitive. Therefore, CRES steel bushings have been installed in the area of the Ground Support Equipment (GSE) attachment positions.

Question: What is the material of engine mount strut?

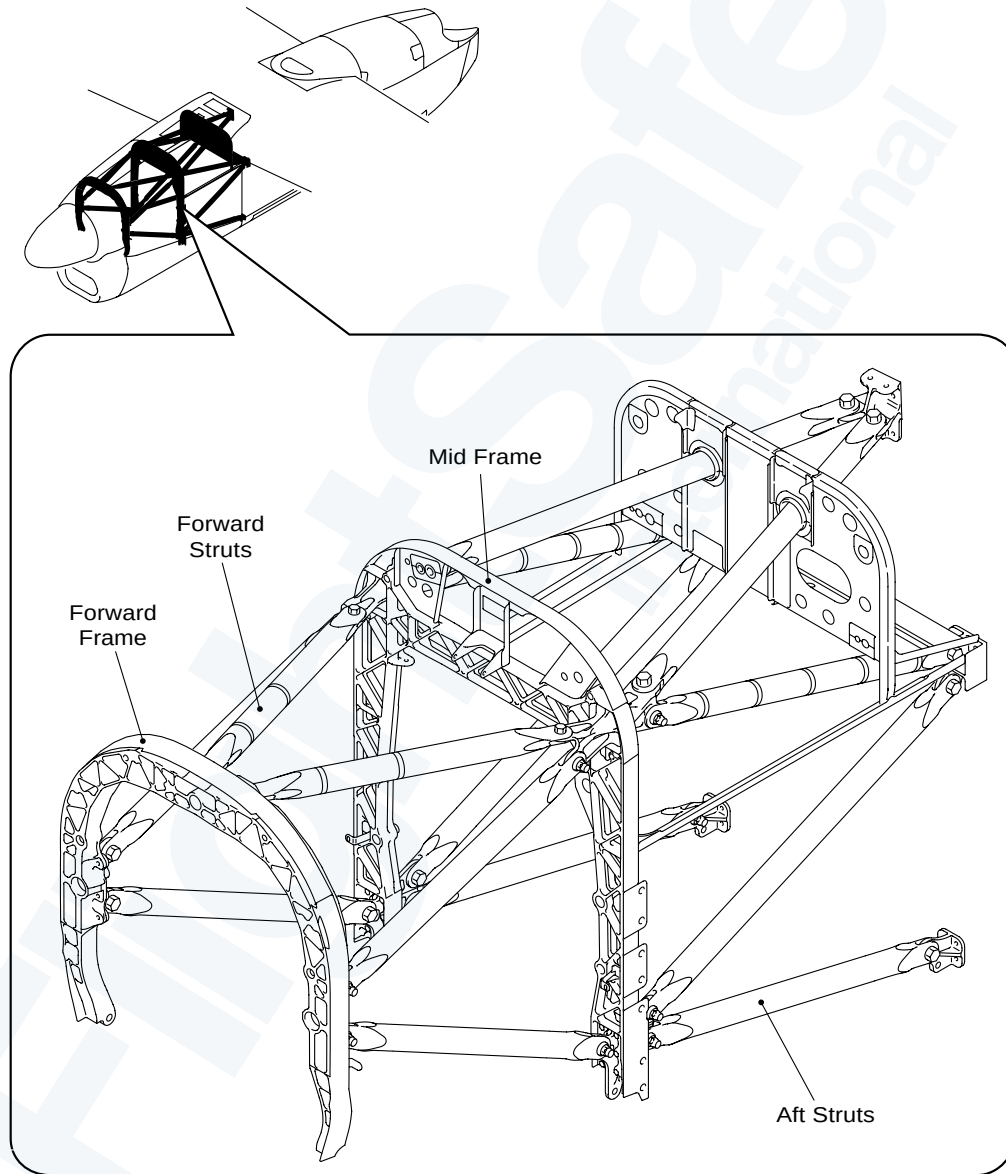
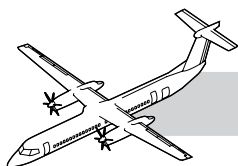


Figure 54-2. Forward Nacelle Structure, Frame and Struts



The mid-frame is a machined titanium component. The frame is attached to the rear engine isolators, distributing the loads from the engine into the engine mount system. It also provides stability for the struts and serves as an attachment point for the lower cowl. Similar to the forward frame, CRES steel bushings have been installed at the GSE attachment points.

The lower cowl is an aerodynamic fairing manufactured from titanium and carbon fibre. It provides an alternative load path in case of failure within the engine mount system. It is attached to the nacelle by hinges and latches.

The lower cowl has three frames located at the major load points. These frames are contoured to match the aerodynamic shape of the cowl. The forward frame is manufactured from two titanium and two aluminium alloy sections. The mid frame is manufactured from three aluminium alloy sections, whilst the aft frame is a single titanium machining.

The external skin and intake duct are moulded in a carbon fibre/nomex honeycomb to form the aerodynamic surface of the cowl.

The firefloor is an assembly of machined and fabricated titanium components. It attaches to the frames and skin and provides additional stiffness.

The air intake is integral to the cowl and is manufactured from a fireproof monolithic laminate.

There are two engine access doors on each side of the nacelle. The doors are of composite construction and hinged along their upper edges.

The forward nacelle fairings comprise the upper forward cowl, the leading edge fairing and the spine cowl.

Forward Nacelle Struts

The space frame strut configuration for the engine mounting structure gives:

- Structural redundancy
- Rotor burst and damage tolerance

- Lateral stiffness
- Low Weight
- Access for maintenance.

The struts are made from machined and forged stainless steel with fork attachments at both ends. The strut fittings are machined titanium construction with provision for bolt attachment to the forward and mid frame positions. The fittings are located at three locations on the aft face of the forward frame, and at the forward and aft face of the mid frame. The fittings are attached to the frames by bolted connection, and the strut bolt connections to the fittings are bushed.

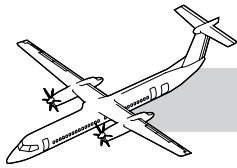
Leading Edge Fairing

Refer to Figure 54-3. Leading Edge Fairing..

The leading edge fairing provides an aerodynamic transition into the wing. It is made from carbon epoxy composite with foam stiffening. Copper mesh on the outer surface gives HIRF and lightning strike protection. The fairing has an access panel for access to the upper area of the zone, and leading edge bay for maintenance.



Figure 54-3. Leading Edge Fairing

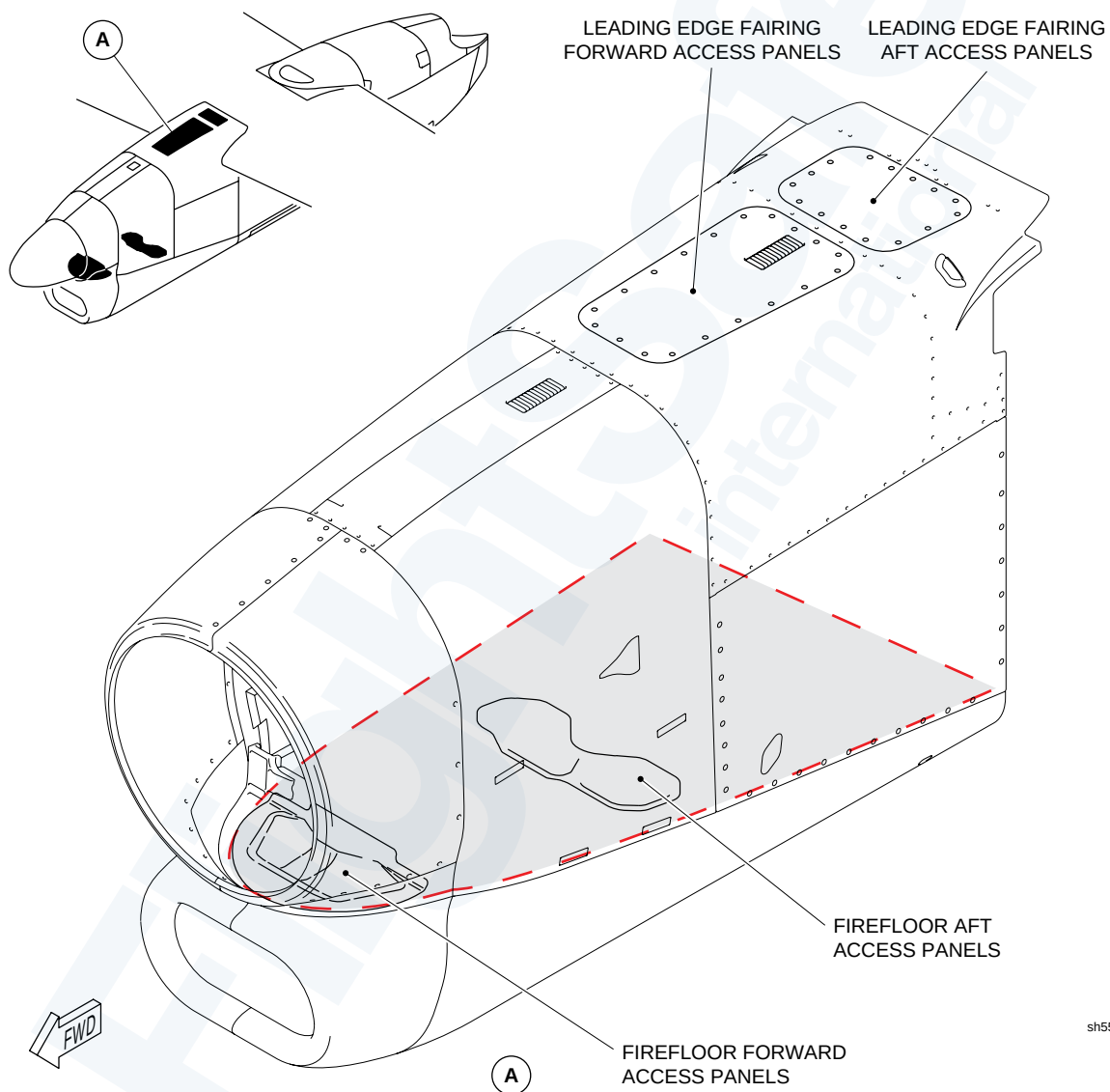


Fire Floor Fwd and Aft Access Panel

Refer to Figure 54-4. Forward Nacelle Access Panels.

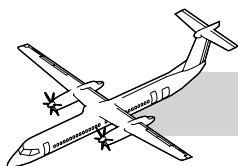
The fire floor forward access panel is installed in the forward nacelle at the forward edge of the lower cowl.

The fire floor aft access panel is installed in the forward nacelle at the middle of the lower cowl.



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Figure 54-4. Forward Nacelle Access Panels



Special Tools and GSE

Prybar

Refer to Figure 54-5. Prybar.

Used to install lower forward cowl.



Figure 54-5. Prybar

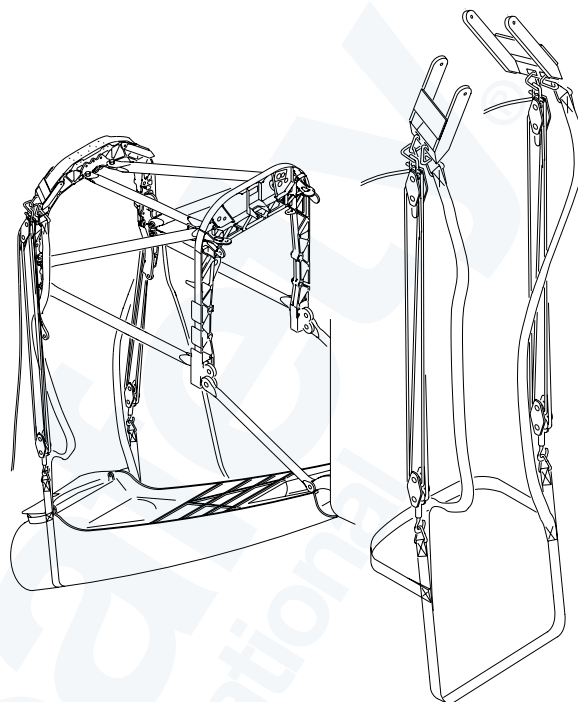


Figure 54-6. Lower Forward Cowl Sling

Lower Forward Cowl Sling and Lanyard

Refer to:

- Figure 54-6. Lower Forward Cowl Sling.
- Figure 54-7. Lower Forward Cowl Lanyard.

The lower cowl lanyard is required to hold the weight of the lower cowl during maintenance procedures. The cowl must be lowered with cowl sling before use.

The lower cowl sling is required to raise and lower the lower cowl during maintenance on removal and installation procedures. After lowering for maintenance, the lanyard can be used to hold cowl in the open position while the sling can be removed and used at the other engine nacelle cowl.

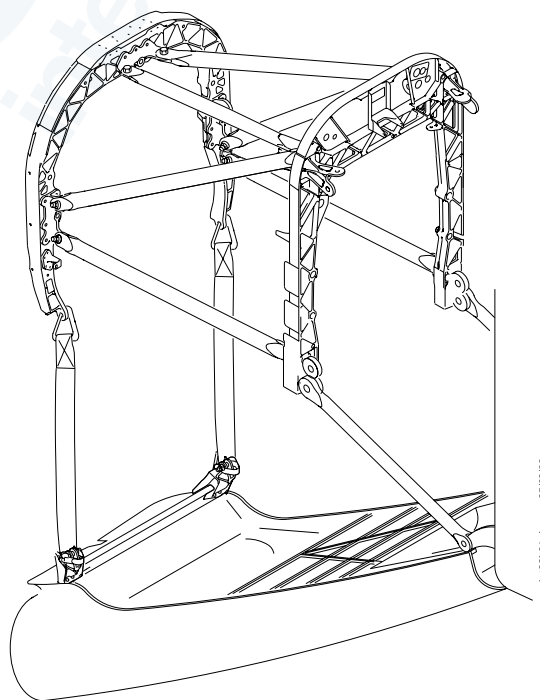
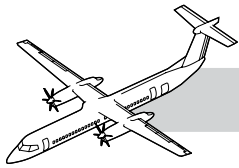


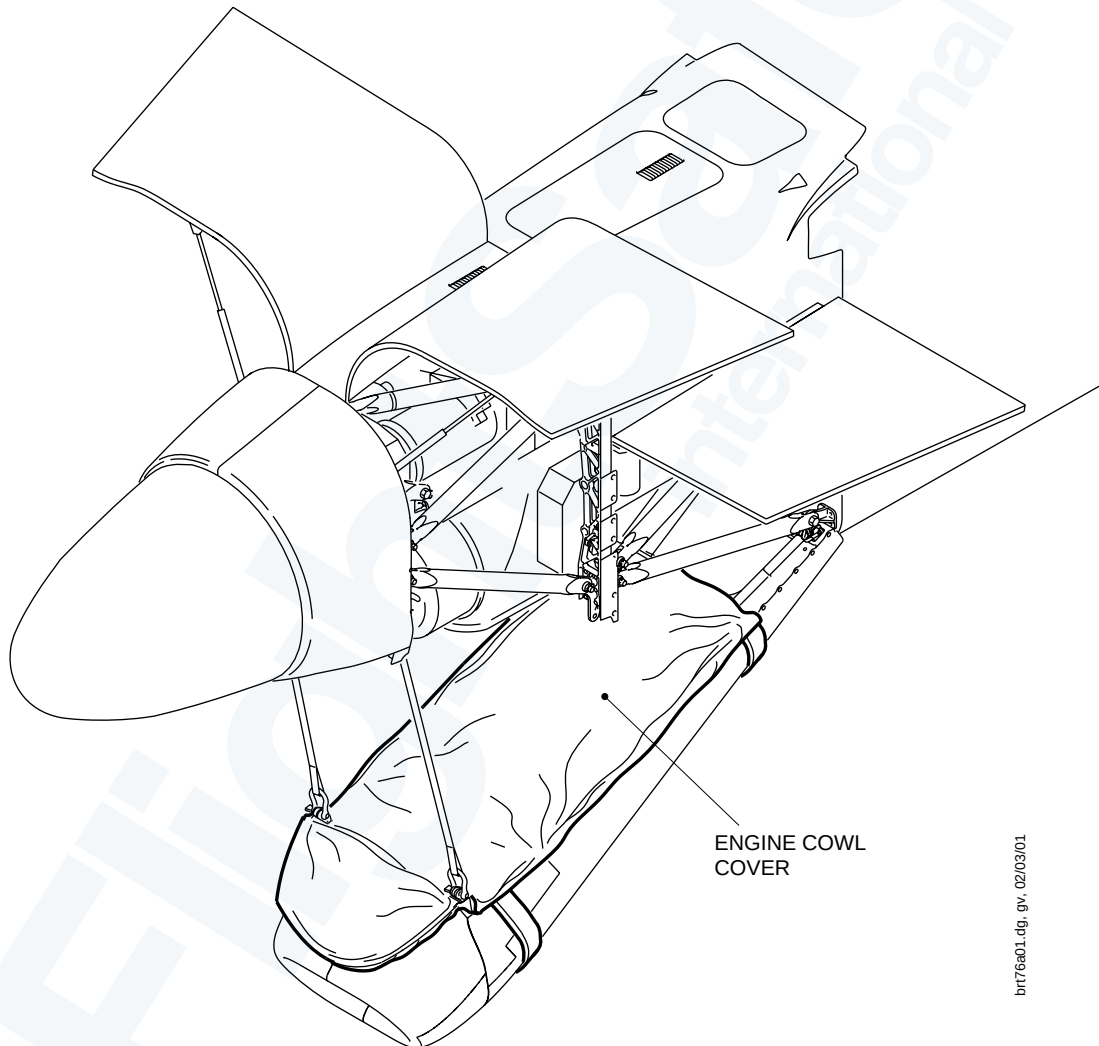
Figure 54-7. Lower Forward Cowl Lanyard



Engine Lower Cowl Cover

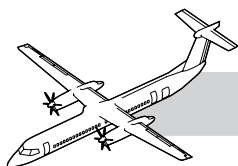
Refer to Figure 54-8. Engine Lower Cowl Cover.

The engine cowl cover prevents unwanted materials from falling into the cowl when it is lowered for engine maintenance. Two covers are required for each aircraft. The cover is manufactured from a material which is resistant to engine and hydraulic fluids. The cover fastens to the cowl by nylon straps.



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Figure 54-8. Engine Lower Cowl Cover



Lower Forward Cowl Hoist

Refer to Figure 54-9. Lower Forward Cowl Hoist.

The lower cowl hoist is required to remove or install the lower cowl from the aircraft to the support/transport stand. The lower forward cowl sling must be used in preparation for use of the hoist.

CAUTION

THE LOWER COWL IS VERY HEAVY. YOU MUST USE THE CORRECT REMOVAL EQUIPMENT WHEN YOU REMOVE THE LOWER COWL. IF YOU DO NOT DO THIS, YOU CAN CAUSE DAMAGE TO THE COWL AND THE NACELLE.

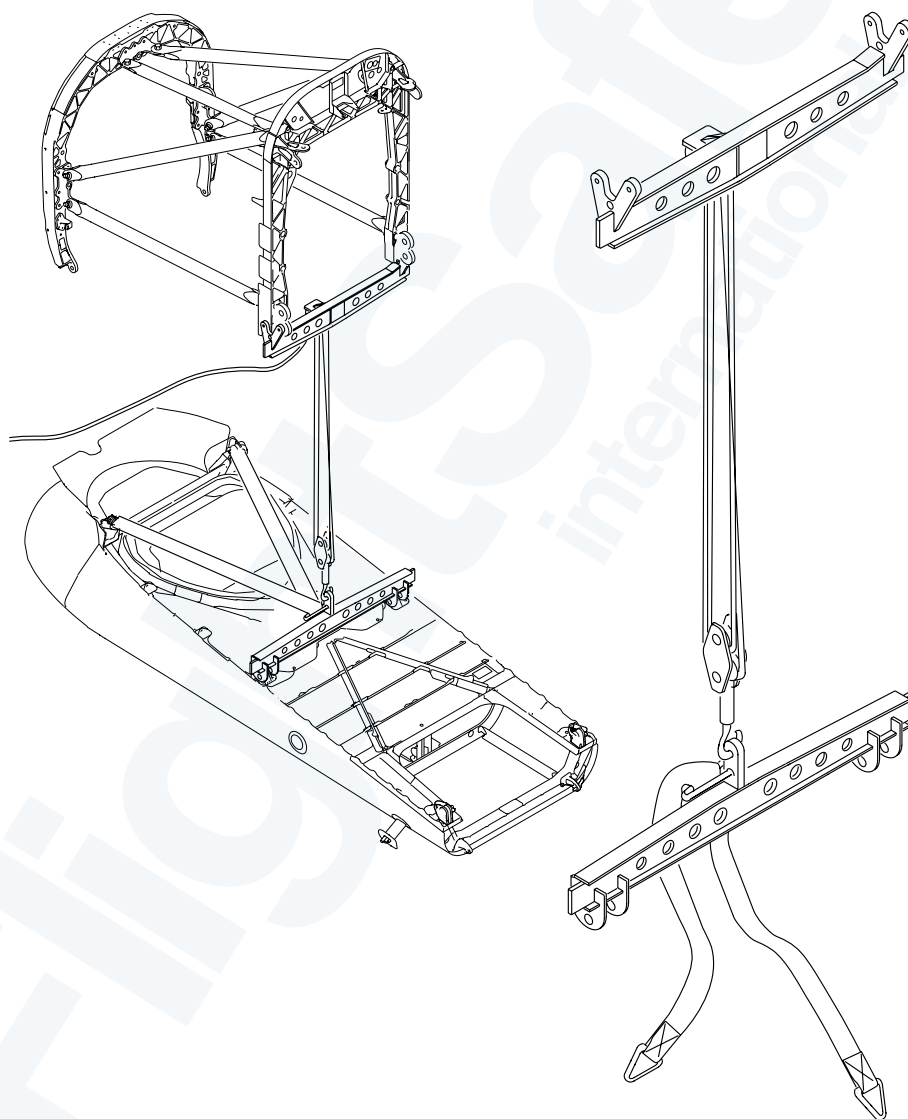
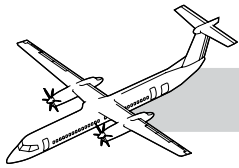


Figure 54-9. Lower Forward Cowl Hoist



Lower Forward Cowl Support/ Transport Stand

NOTE

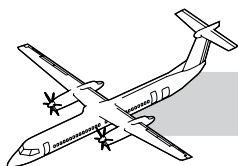
Refer to Figure 54-10. Stand Lower Cowl.

The stand is required to hold or carry the lower forward cowl during removal or installation procedures.

Made from steel frame and wooden supports, padded with rubber foam to protect cowl.



Figure 54-10. Stand Lower Cowl



Center Nacelle Structure

Refer to Figure 54-11. Center Nacelle Structure.

The center nacelle structure is located between nacelle stations 121.230 and 210.000 (omitting the forward upper section of the aft nacelle) and comprises the components that follow:

- A-Frame
- Two main side frames
- Skins
- A rear bulkhead
- Main landing gear attachments
- Nacelle-to-wing attachments.

The A-Frame is machined from a solid aluminium alloy billet and attaches to the wing front spar, the side frames, the engine mounting struts and the MLG drag-strut and side-braces.

Machined upper and lower longerons are attached to machined and sheet metal aluminium frames, closed with aluminium alloy outer skins. The two-piece upper longeron is separated by a machined fitting which links the landing gear yoke to the nacelle and the nacelle to the rear wing spar. A “coat hanger” brace links the two side frames and the rear spar.

The lower skin on the center section comprises four landing gear doors, manufactured from a carbon fibre/nomex honeycomb assembly.

The rear bulkhead of the center nacelle is a machined aluminium alloy frame with a ‘V’ shaped cut-out to provide access for the landing gear.

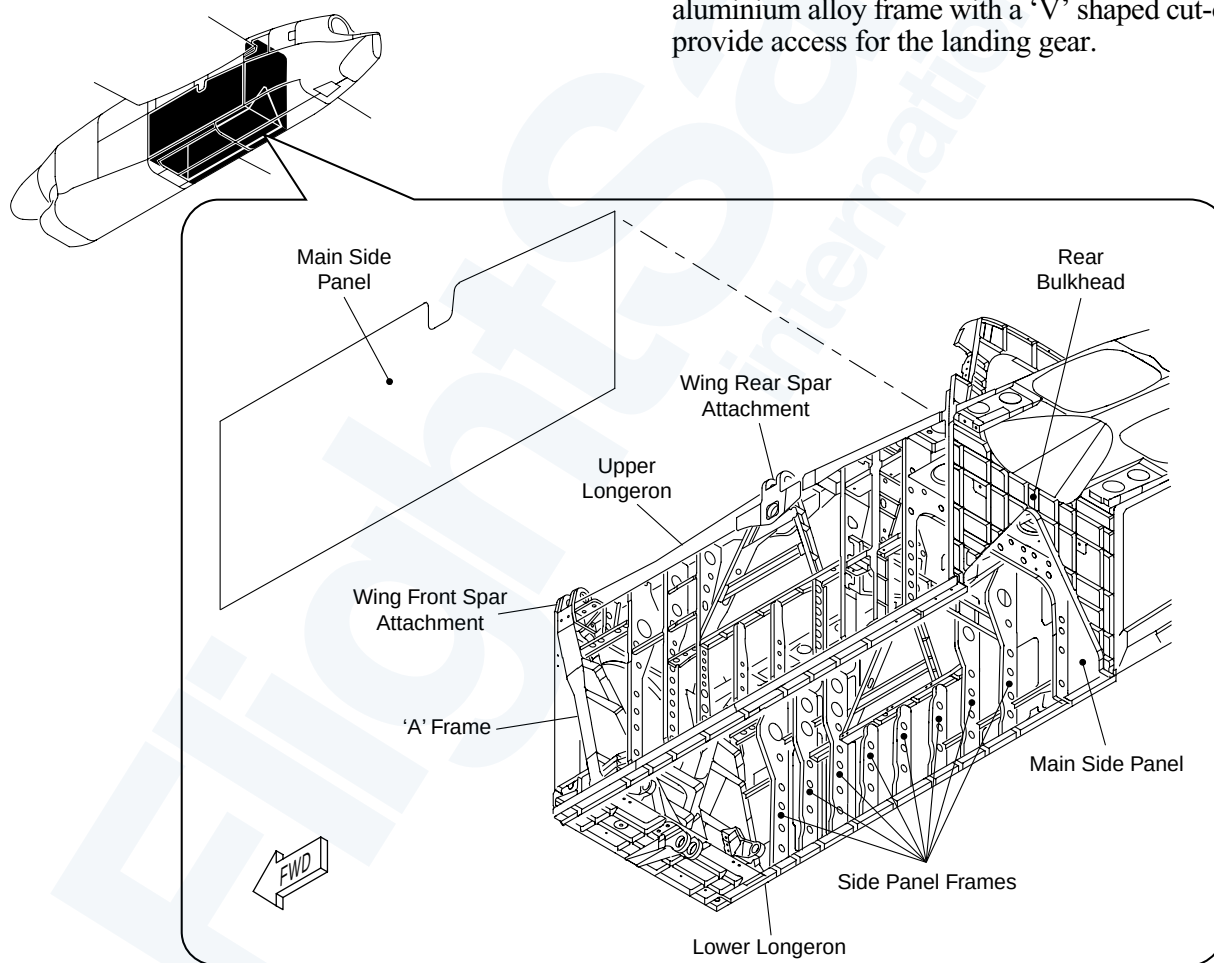
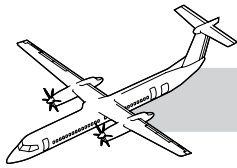


Figure 54-11. Center Nacelle Structure



Nacelle Access Panels

Refer to Figure 54-12. Center Nacelle Access Panels.

The access panels can be removed to do inspection and maintenance tasks.

These access panels are machined from solid aluminium alloy. The panels are removed to gain access to various attachments.

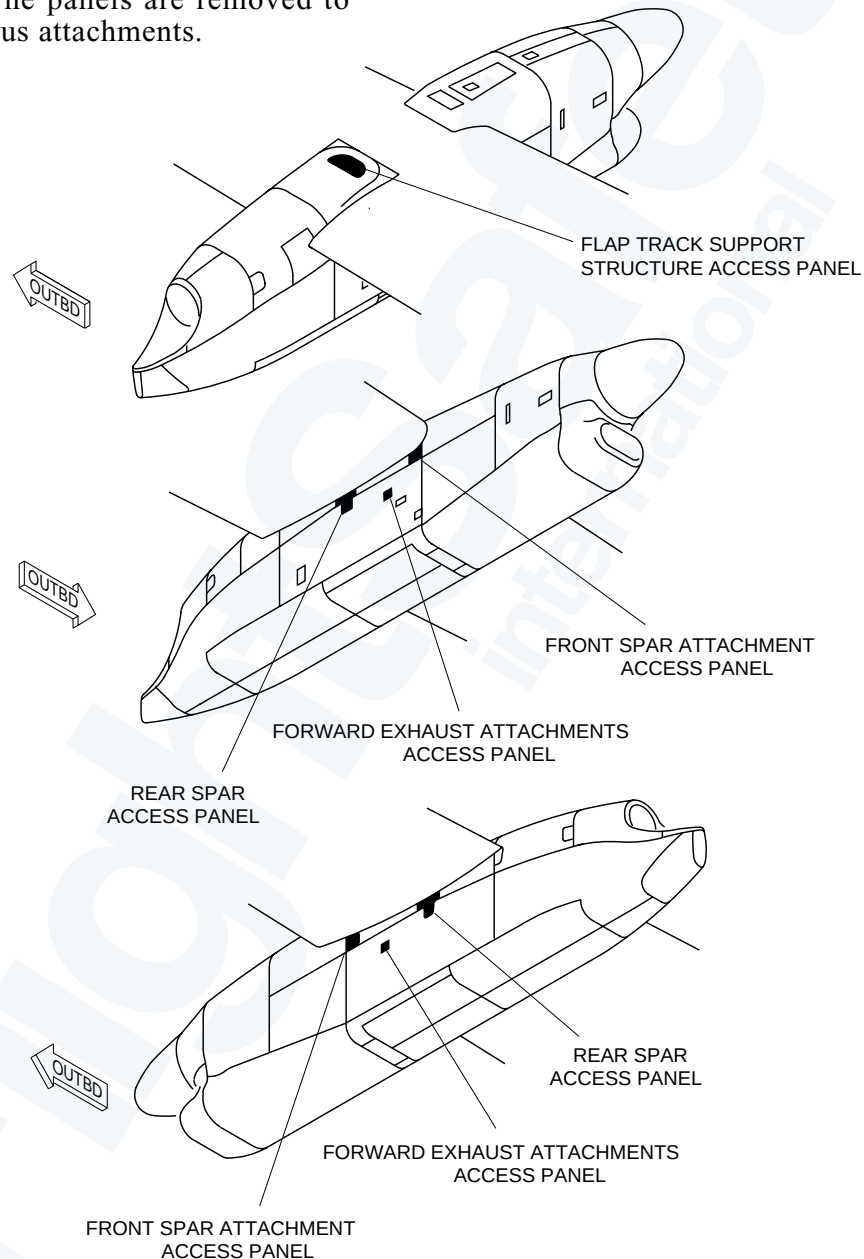
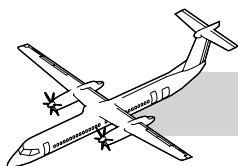


Figure 54-12. Center Nacelle Access Panels



54-21-31 Back-Up Fitting Access Panels

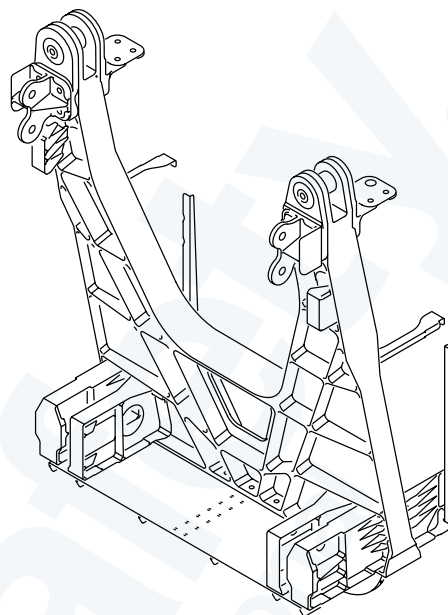
Refer to:

- Figure 54-13. A - Frame - Structural Identification.
- Figure 54-14. Back Up Access Panel (415BL/425BL) - Removal/Installation.

Allows access to the backup fittings for Inspection and Replacement.

NOTE

These are sealed panels and must be installed wet with Sealant. Don't forget to use release agent.



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Figure 54-13. A - Frame - Structural Identification

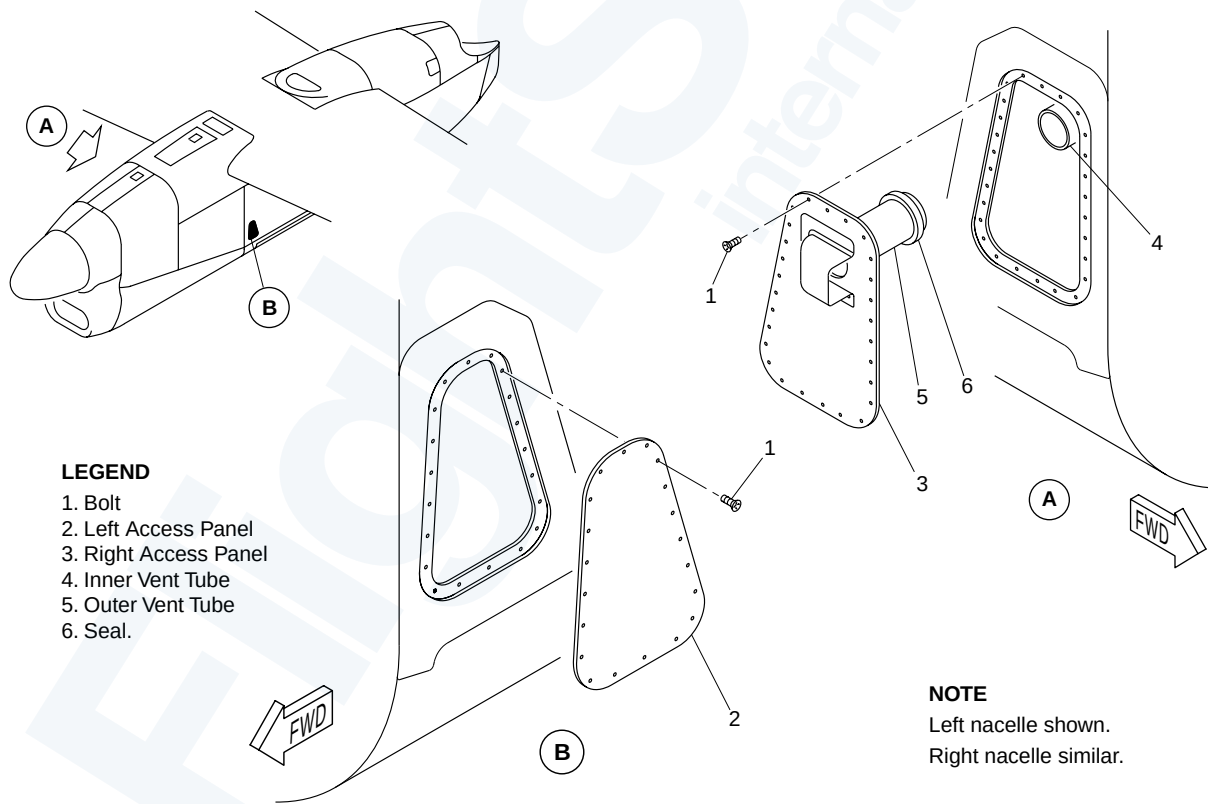
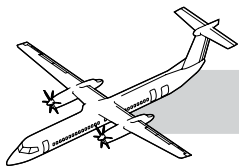


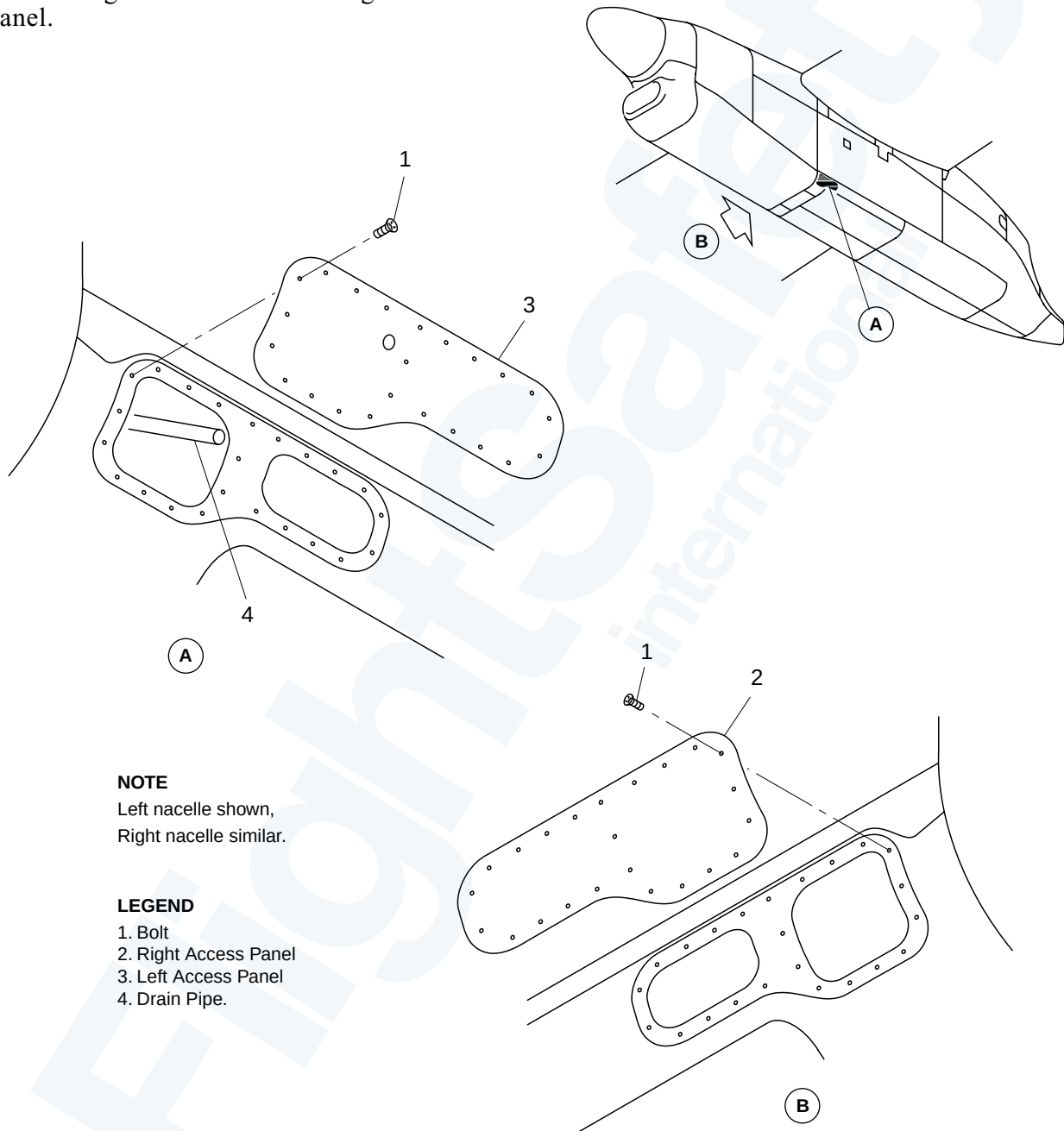
Figure 54-14. Back Up Access Panel (415BL/425BL) - Removal/Installation



54-21-36 Lower Longeron Access Panels

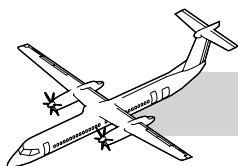
Allows access to the Lower Longerons for Inspection and Replacement.

Refer to Figure 54-15. Lower Longeron Access Panel.



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Figure 54-15. Lower Longeron Access Panel



Wing Front and Rear Spar Attachments

Refer to Figure 54-16. Coat Hanger Assemblies - Structural Identification.

The wing rear spar attachments to the nacelle have two fittings machined from aluminium alloy. The attachments supply nacelle to wing, main landing gear yoke and coat hanger interfaces. The attachment fittings are attached to the longerons and mounted off the rear spar. The coat hanger is also attached to the rear spar at its mid-point.

The wing front spar attachment is through the A frame integral lugs, with forward and aft, steel fittings, for multi-load path.

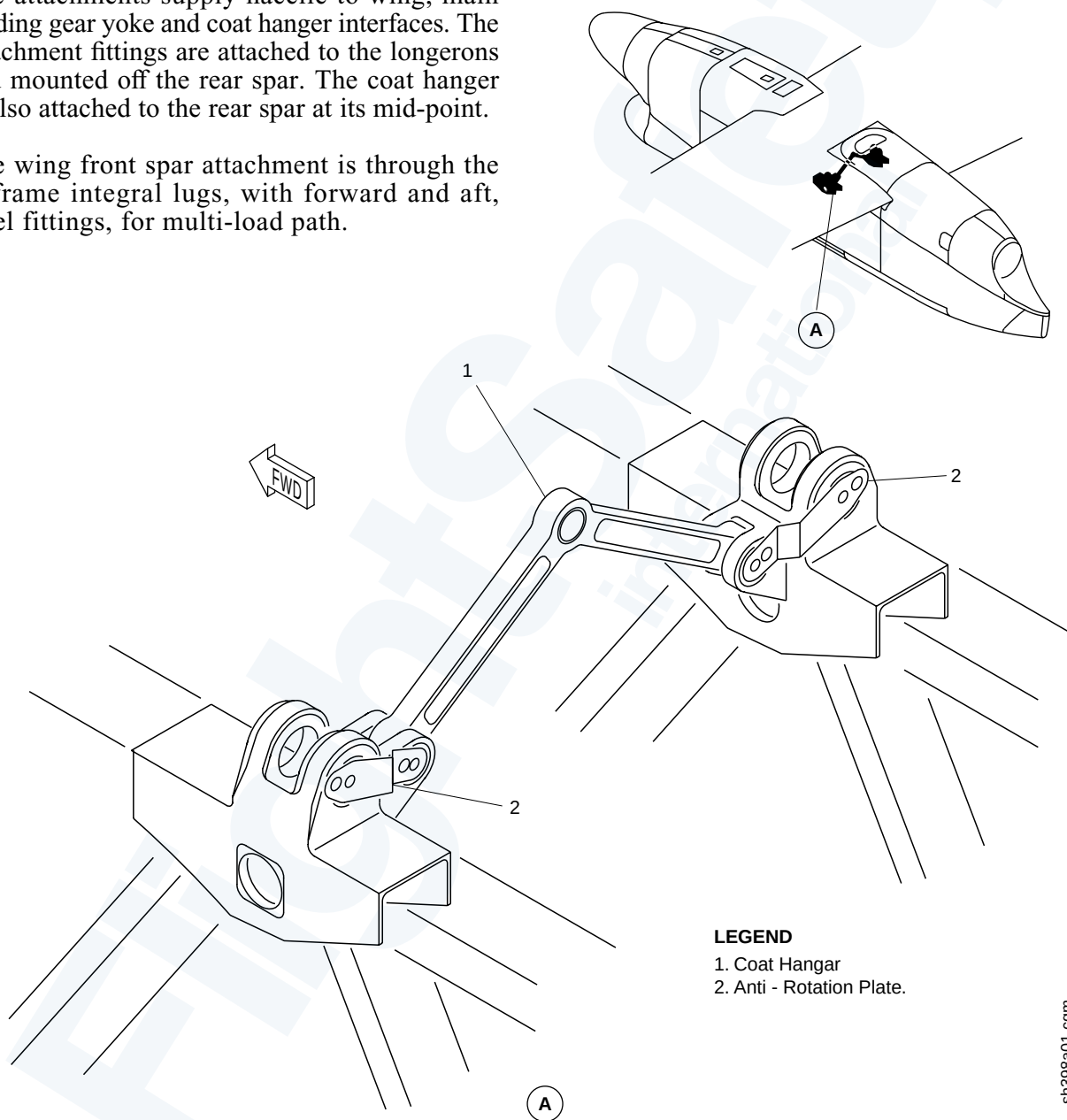


Figure 54-16. Coat Hanger Assemblies - Structural Identification

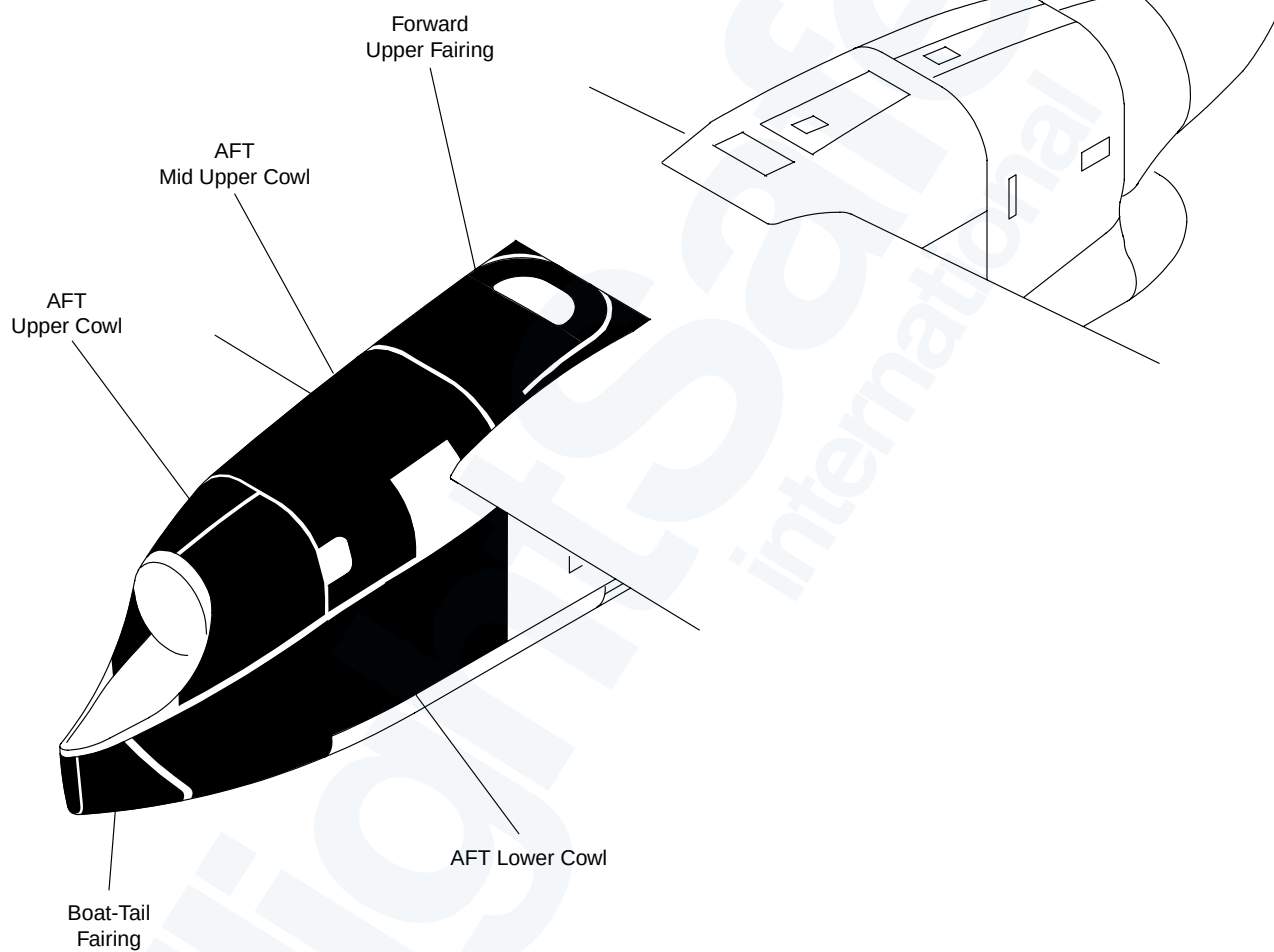
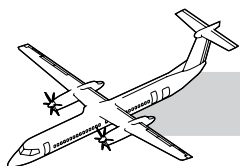
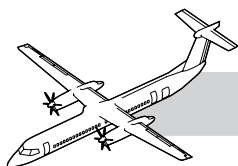


Figure 54-17. Aft Nacelle Structure (1 of 2)



Aft Nacelle Structure

NOTE

Refer to Figure 54-17. Aft Nacelle Structure (1 of 2).

The aft nacelle structure is located between nacelle stations 171.65 to 315.063 and includes the components that follow:

- An aft lower section
- A forward-upper fairing (containing the flap track mounting structure)
- A mid-upper fairing
- An aft-upper cowl (which provides an outlet for exhaust gases).

The side panels of the aft lower section are constructed from a carbon fibre/nomex honeycomb assembly. The right nacelle panels have a cut-out to allow the inclusion of a refuel access door.

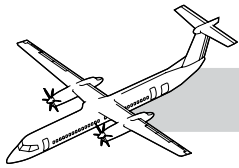
The boat tail fairing is a carbon fibre/nomex honeycomb assembly closed at its aft end by a small sheet aluminium alloy fairing.

The top of the aft lower nacelle is closed by an aluminium alloy heatshield containing wheel blisters at its forward end.

The forward upper fairing contains the No.2 and No.3 flap tracks mounted on machined aluminium frames fabricated into a box covered by carbon fibre skin.

The mid upper fairing is a 'U' shaped carbon fibre/nomex honeycomb fairing.

The aft-upper cowl is made from two aluminium alloy skin halves leaving an opening for the exhaust duct exit.



54-31-00 Aft Nacelle Access Panels

Refer to Figure 54-18. Aft Nacelle Structure
(2 of 2).

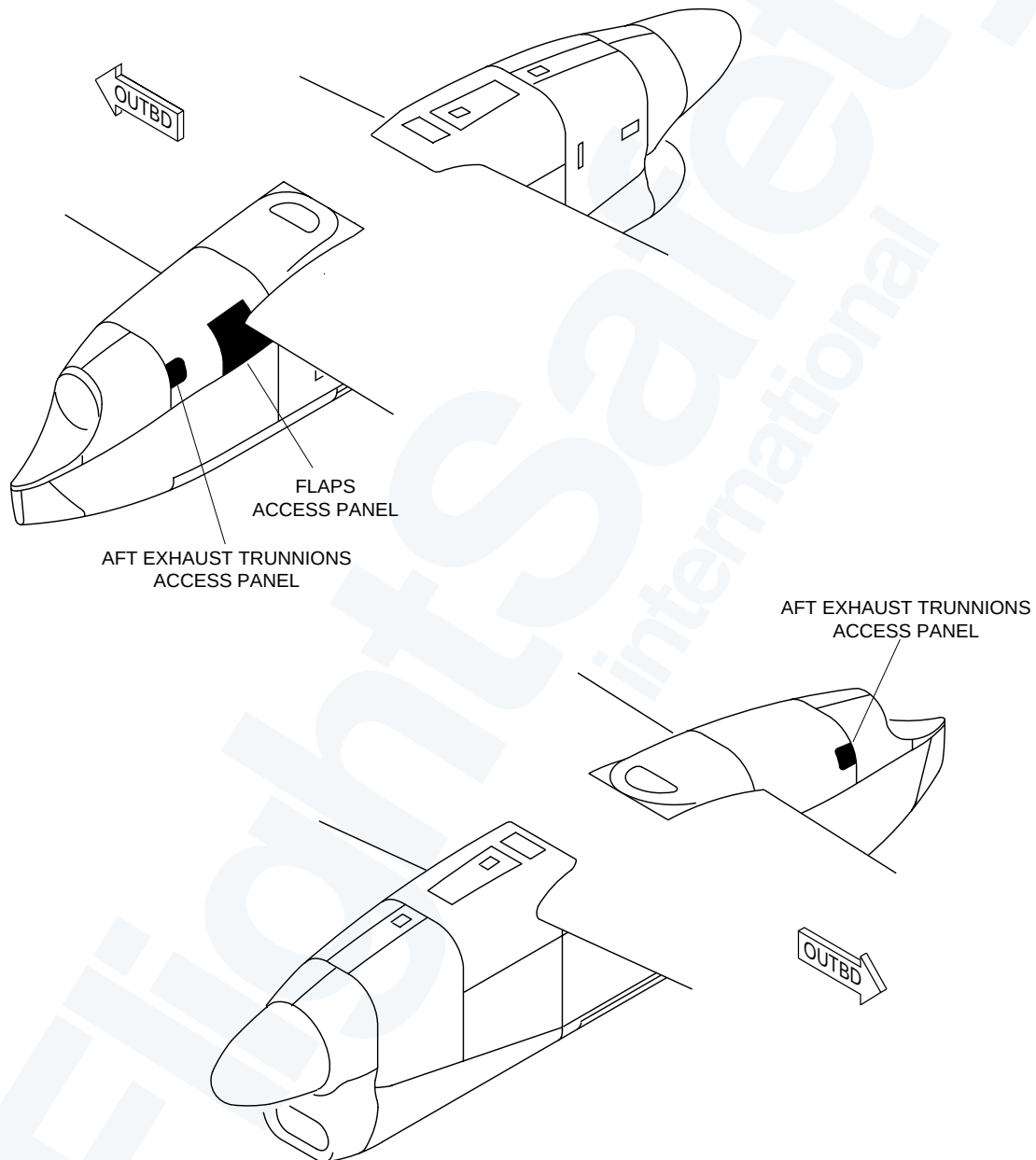
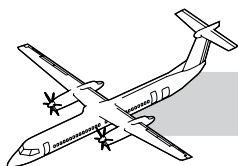


Figure 54-18. Aft Nacelle Structure (2 of 2)

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54-32-16 Aft Upper-Nacelle Seals

Refer to Figure 54-19. Aft Upper Nacelle Seals - Removal/Installation.

If the airflow moves from the lower to the upper surface of the wing trailing edge it creates vortices and drag.

To keep this from happening the manufacturer has installed seals on each side of the aft upper skin assembly of the nacelles.

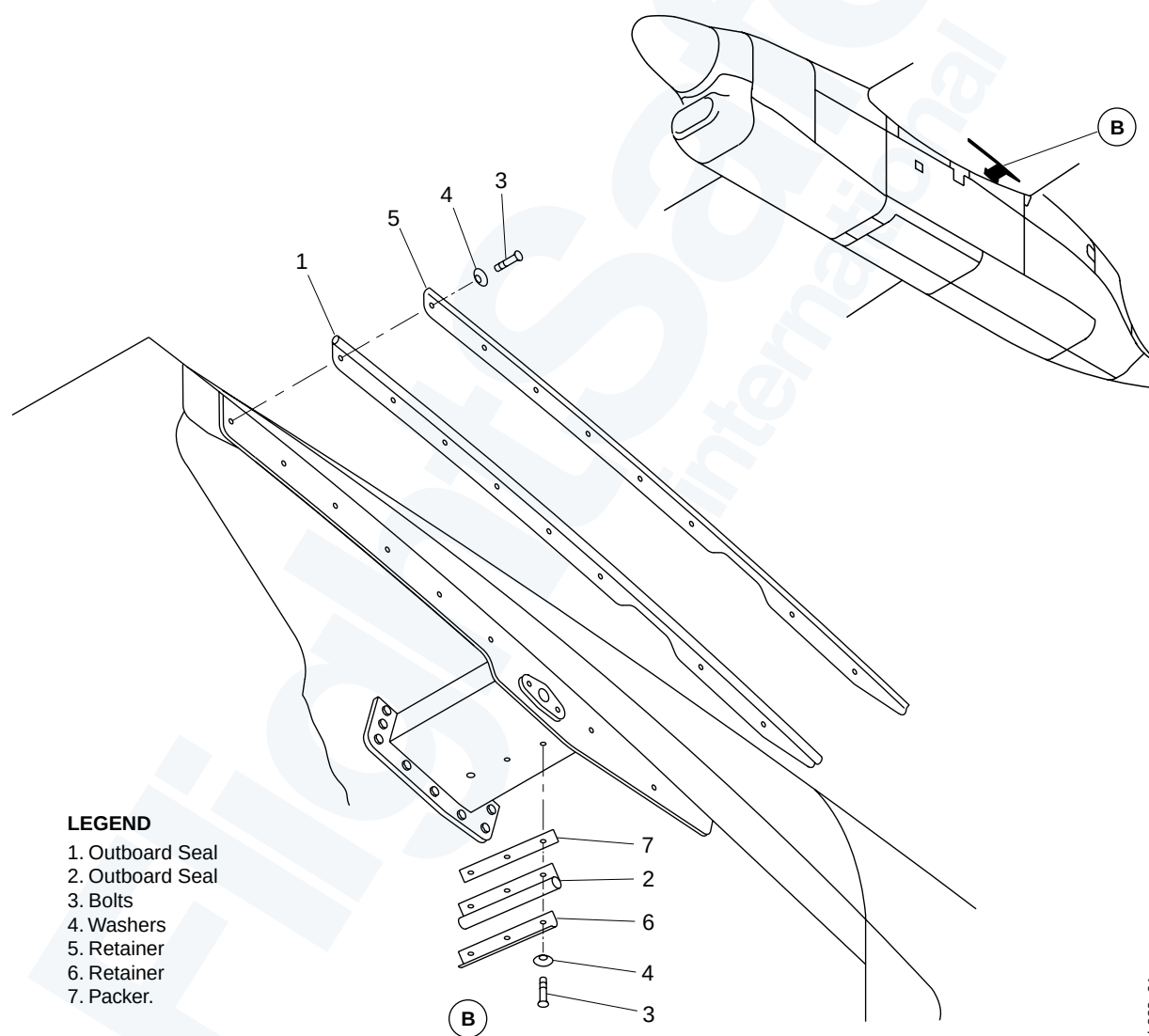
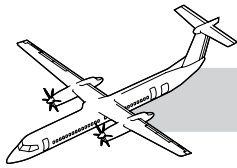
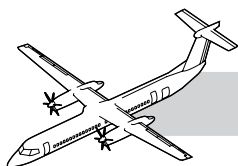


Figure 54-19. Aft Upper Nacelle Seals - Removal/Installation



54-00-00 SPECIAL TOOL & TEST EQUIPMENT

- GSB5411003 Cover - Engine Lower Cowl (Series 400)
- ACS28004 Sling - Lower Forward Cowl (Series 400)
- ACS28009 Hoist - Lower Forward Cowl (Series 400)
- ACS28005 Stand - Lower Forward Cowl Support/Transport (Series 400)
- GSB5400001 Bar - Pry
- Commercial Weighing Scale, triple beam balance type, weighing tolerance within ± 0.5 grams
- GSB2000001 Borescope - 110 Volt, 60 Hz (if necessary)
- GSB2000015 Borescope - 220 Volt, 50 Hz (if necessary)
- GSB0700025 (or equivalent) Engine strap
- Commercially available Load cell (able to measure a 2500 lbs (1134 kg) load)
- GSB0700024 Stand - Tail Support
- Commercially available Calibration Test Standard for HFEC inspection, P/N SRS 0824A (NDT Engineering Corporation)
- Commercially Available Calibration Test Standard (CTS) for BHEC inspection, "Navy Universal Eddy Current Reference Standard", P/N NRK-3A, S/N 2621 (PH Tool)
- Commercially available Load cell (able to measure a 2500 lbs (1134 kg) load).



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